

Auditing Engagement Incentives in the Kidfluencer Ecosystem: A Multimodal Weak Supervision Approach

Manus AI

Abstract

The rapid rise of “kidfluencers” on YouTube has raised profound ethical concerns regarding child digital labor and exploitation. While emerging legislation attempts to regulate this ecosystem, empirical evidence on the relationship between child exploitation and engagement metrics remains scarce due to the challenge of operationalizing and scaling exploitation measurements. This study presents a multimodal AI audit of 4,685 videos across 79 kidfluencer channels (sampled from a dataset of 58,965 videos), utilizing a weak supervision approach (Snorkel) to detect exploitation signals without requiring large-scale manually labeled ground truth. We aggregate 18 noisy labeling functions—including LLM-based classification of titles across six literature-grounded dimensions, rule-based heuristics, and computer vision analysis of thumbnail distress signals—to assign a probabilistic exploitation score to each video.

Our findings reveal a significant **engagement premium associated with performative labor and manufactured conflict**. Overall exploitation scores correlate significantly with view counts (Spearman $\rho = 0.159$, $p < 10^{-28}$). Within-channel analyses show that performative content is associated with a median view boost of +42.0% (FDR-corrected $p < 0.001$), while narrative conflict is associated with a median boost of +32.0% ($p = 0.002$). These effects hold in robustness checks comparing videos published within the same year ($p = 0.0018$). Unlike previous qualitative assumptions, we find that commercial content (product placement) has no significant effect on viewership (+3.7%, $p = 0.560$), suggesting that the platform ecosystem associates higher viewership with the commodification of the child’s identity and labor rather than traditional advertising. These findings challenge policy frameworks

focused solely on financial trusts, demonstrating that engagement metrics systematically reward the intensive, performative labor of children.

1. Introduction

The commercialization of childhood through digital platforms has created a lucrative “kidfluencer” economy in which children are featured in YouTube videos—unboxing toys, participating in challenges, performing scripted roleplay, and documenting their daily lives [11]. This phenomenon has sparked intense debate regarding the ethics of child digital labor, the psychological impact of public exposure, and the blurring of lines between play and work, often termed “playbour” [16].

Recent legislative efforts, such as the Illinois PA 103-0556 (2024) [29], aim to protect child creators by mandating financial trusts and limiting working hours. However, these regulations treat kidfluencing as a traditional labor market, often failing to address how platform ecosystems actively shape content creation [31]. A fundamental question remains: **How do engagement metrics associate with specific dimensions of child exploitation?**

Previous research on the kidfluencer ecosystem has largely relied on qualitative content analysis or small-scale manual coding [13, 14]. While valuable, these approaches cannot scale to audit the massive volume of content generated. Conversely, purely computational approaches often struggle to operationalize complex, nuanced concepts like “exploitation” or “performativity” without expensive, large-scale human annotation. Furthermore, prior audits of platform algorithms [2, 3] have often focused on political content rather than child safety [8].

This study bridges this gap by deploying a **multimodal weak supervision pipeline** to conduct a large-scale observational audit. We ground our definition of exploitation in the UN Convention on the Rights of the Child (UNCRC) and recent theoretical frameworks [11, 13], operationalizing six specific dimensions: performative labor, emotional bait, narrative conflict, challenge formats, commercial content, and privacy violations.

Crucially, because we rely on observational data via the YouTube API, we cannot directly observe the recommendation algorithm’s internal scoring [5, 34]. Instead, we measure the **engagement premium**—the association between exploitative content

dimensions and view counts—which serves as a proxy for the incentive structures shaping creator behavior.

Our core research questions are:

- **RQ1:** Can a weak supervision framework effectively synthesize multimodal signals (text, LLM classifications, and computer vision) to measure kidfluencer exploitation at scale?
- **RQ2:** Are specific dimensions of exploitation (e.g., performative labor vs. privacy violations) associated with higher view counts (an “engagement premium”)?
- **RQ3:** Does this engagement premium persist *within* channels and when controlling for video age, indicating a structural correlation rather than merely a channel-popularity effect?

2. Related Work

2.1 The Kidfluencer Economy and Exploitation Frameworks

The kidfluencer economy relies on the continuous documentation of children’s private lives and their participation in scripted entertainment. Clark and Jno-Charles [11] propose analyzing this phenomenon through the lens of the UNCRC, identifying fundamental threats to children’s rights: economic exploitation, exposure to harm, and restriction of authentic expression. Divon et al. [13] further describe how children are transformed into “concealed commodities” through practices like transactional play.

Other literature highlights the privacy risks of “sharenting” [17, 18], where parents overshare children’s lives online, potentially leading to emotional neglect [19]. In extreme cases, such as the “Elsagate” phenomenon, platforms have struggled to moderate inappropriate content targeting toddlers [8, 9, 10]. Building on these frameworks, we operationalize exploitation not merely as overt abuse, but as the intensive, performative labor required to maintain engagement.

2.2 Algorithmic Auditing and Engagement Metrics

Algorithmic auditing investigates platform behavior without direct access to proprietary code [35, 36]. Studies have audited YouTube and Twitter for political

radicalization and amplification [1, 2, 3, 4], often using sock puppets [6] or observational engagement data [5]. Since direct recommendation rates are hidden, researchers often use engagement metrics (views, likes) as proxies for algorithmic reach [32, 33, 34]. We adopt this observational approach, focusing on the “engagement premium” associated with specific content types.

2.3 Weak Supervision and LLM Content Analysis

Traditional machine learning requires massive labeled datasets, which are difficult to obtain for subjective concepts. Weak supervision frameworks like Snorkel [20, 21, 22] allow researchers to encode domain knowledge as noisy heuristic rules (Labeling Functions) to generate probabilistic labels [23]. Recently, Large Language Models (LLMs) have shown promise in zero-shot content moderation and annotation [24, 25, 26]. We combine LLMs with Snorkel to scale our exploitation analysis.

3. Methodology

3.1 Data Collection and Sampling

We collected metadata for 58,965 videos from 79 family and kidfluencer YouTube channels using the YouTube Data API. Channels were selected based on prior literature and popular influencer lists, covering a spectrum of channel sizes and target audiences. Animated channels were strictly excluded; all selected channels feature real children.

To manage computational costs while maintaining representativeness, we employed a stratified sampling strategy. For each of the 79 channels, we stratified videos into terciles based on view counts (high, medium, low) and randomly sampled up to 20 videos per tercile, resulting in a final stratified sample of 4,685 videos.

3.2 Exploitation Dimensions

Based on Clark and Jno-Charles [11] and Divon et al. [13], we defined six exploitation dimensions:

1. **Performative Labor:** Child performing scripted/planned content for the camera.
2. **Emotional Bait:** Using the child’s exaggerated emotions for clickbait.

3. **Narrative Conflict:** Manufactured drama or conflict involving the child.
4. **Challenge Format:** High-effort competition formats requiring extended labor.
5. **Commercial Content:** Child used explicitly for product placement/unboxing.
6. **Privacy Violation:** Exposure of the child’s private, vulnerable, or medical moments.

3.3 Multimodal Weak Supervision Pipeline

We implemented a Snorkel-based weak supervision pipeline utilizing 18 Labeling Functions (LFs) across three modalities:

LLM-Based LFs (6): We deployed GPT-4.1-mini to classify video titles along the six dimensions defined above. Each classification served as a distinct LF voting for or against exploitation.

Rule-Based LFs (9): We developed heuristics based on title metadata, including all-caps ratios, excessive exclamation marks, and keyword dictionaries targeting conflict, challenges, pranks, and organic family events.

Computer Vision LFs (3): We processed thumbnails using OpenCV to extract color saturation, as hyper-saturated thumbnails indicate visual manipulation strategies. Additionally, we deployed the GPT-4.1-mini Vision API on a subsample to detect child distress and assess visual exploitation concern.

The Snorkel Label Model aggregated these 18 noisy signals to assign a continuous probabilistic **Exploitation Score** $\in [0, 1]$ to each video.

4. Results

4.1 Pipeline Performance and Dimension Prevalence

The Label Model successfully aggregated the multimodal signals, predicting 24.1% of the sample as exploitative ($P(\text{exploit}) > 0.5$) and 75.9% as non-exploitative. LLM classification revealed that **performative labor** is the most prevalent dimension (17.4% of videos), followed by challenge formats (12.8%), emotional bait (11.4%), and narrative conflict (9.0%). Direct privacy violations (3.2%) and explicit commercial content (3.7%) were less common.

 Figure 1: Exploitation Dimension Prevalence and Score Distribution

Figure 1: (a) Distribution of the probabilistic exploitation score generated by the weak supervision model. (b) Correlation between exploitation score and view count.

 Figure 5: Dimension Prevalence

Figure 2: Prevalence of literature-grounded exploitation dimensions across the stratified sample.

4.2 The Engagement Premium (RQ2 & RQ3)

We found a highly significant positive correlation between a video's overall Exploitation Score and its view count (Spearman $\rho = 0.159, p < 10^{-28}$).

To determine if this association is a structural platform correlation rather than merely an artifact of highly exploitative channels being more popular overall, we conducted a **within-channel analysis**. For each dimension, we compared the median views of videos exhibiting that dimension against videos from the *same channel* lacking it. To account for multiple comparisons, we applied False Discovery Rate (FDR) correction.

 Figure 2: Within-Channel View Boost by Dimension

Figure 3: Mean within-channel view boost by exploitation dimension.

The results reveal a clear engagement premium structure:

- **Performative Labor** is associated with a median within-channel boost of +42.0% (mean +61.7%, FDR $p < 0.001$).
- **Narrative Conflict** is associated with a median boost of +32.0% (mean +48.6%, FDR $p = 0.002$).
- **Challenge Formats** are associated with a median boost of +14.8% (mean +62.5%, FDR $p = 0.007$).
- **Emotional Bait** is associated with a median boost of +13.3% (mean +79.5%, FDR $p = 0.007$).

Notably, **Commercial Content** (explicit product placement/unboxing) showed no significant effect on viewership (median +3.7%, mean +6.2%, FDR $p = 0.560$). Privacy violations also showed no significant effect (median -7.4%, FDR $p = 0.114$).

4.3 Robustness Checks

To ensure our findings were not merely artifacts of video age (older videos accumulating more views), we conducted a **same-year within-channel comparison**. By matching exploitative and non-exploitative videos published by the same channel in the same year (26 channel-year groups), we found the engagement premium holds robustly: high-exploitation videos received a median boost of +26.2% over their same-year, same-channel counterparts (Wilcoxon $p = 0.0018$).

A secondary check using views-per-day was underpowered due to missing `publishedAt` metadata in our historical dataset (only 21 channels retained sufficient data), yielding non-significant results. However, the strong same-year findings confirm the effect is not mechanically driven by video age.

4.4 Target Audience as a Moderating Variable

We hypothesized that the target audience moderates the engagement premium. We classified channels into “Child Audience” (11 channels: animated content, toy play) and “Teen/Adult Audience” (66 channels: family vlogs, challenges).

 Figure 4: Audience Moderation

Figure 4: Within-channel exploitation premium by target audience.

For **Teen/Adult-audience channels**, high-exploitation content is associated with a substantial premium (median boost +16.4%). Conversely, for **Child-audience channels**, high-exploitation content is associated with a penalty (median boost −14.7%). This difference showed a moderating trend but was not statistically significant (Mann-Whitney $p = 0.115$), likely due to the small sample size of toddler channels ($n = 11$).

5. Discussion

5.1 The “Performativity Premium”

Our findings provide large-scale empirical evidence for a “performativity premium” in the kidfluencer ecosystem. Engagement metrics are systematically associated with

content that requires children to engage in intensive, performative labor (challenges, scripted conflict, emotional bait) over organic family documentation.

Interestingly, traditional commercial content does not enjoy this premium. This suggests a shift in the kidfluencer economy: the most successful strategy is not to use the child to sell a physical toy, but to make the child’s labor, emotions, and manufactured drama the product itself. This aligns with Divon et al.’s [13] concept of “transactional childhood.”

5.2 Methodological Contributions

This study demonstrates the efficacy of weak supervision for large-scale observational auditing. By combining LLM capabilities, computer vision, and rule-based heuristics within a Snorkel framework, we successfully scaled the operationalization of complex ethical concepts without requiring massive manual ground truth, offering a blueprint for future audits of subjective content moderation issues.

5.3 Limitations and Future Work

This study has several limitations. **First, the probabilistic labels generated by the Snorkel pipeline have not yet been validated against a manually annotated ground-truth dataset.** While weak supervision models the internal agreement of heuristics, its accuracy relative to human judgment remains unverified. An annotation sheet has been prepared for future human validation.

Second, our analysis relies on observational data. We measure *engagement metrics* (views), which are proxies for algorithmic reach, but we cannot claim a direct causal link to YouTube’s internal recommendation weights [34]. Third, our LLM classification primarily analyzed video titles and thumbnails; future work should incorporate full video transcripts and duration data. Finally, our audience moderation analysis was underpowered.

6. Ethics Statement

This research utilizes publicly available, observational data from YouTube via the official API. The study was deemed IRB-exempt as it involves no direct interaction with human subjects or minors, and analyzes public figures in the digital economy. We

report aggregate statistics to protect the privacy of specific children featured in the channels.

7. Conclusion

As the kidfluencer economy matures, regulatory focus must expand beyond financial compensation to address the structural forces shaping content creation. Our multimodal audit of 4,685 videos demonstrates that engagement metrics are significantly associated with content featuring child performative labor, narrative conflict, and emotional bait. By highlighting the “performativity premium,” this study underscores the need for platform-level interventions that disincentivize the commodification of child labor and stress.

References

- [1] Ribeiro, M. H., et al. (2020). Auditing radicalization pathways on YouTube. *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency*, 131–141.
- [2] Huszár, F., et al. (2022). Algorithmic amplification of politics on Twitter. *Proceedings of the National Academy of Sciences*, 119(1).
- [3] Haroon, M., et al. (2023). Auditing YouTube’s recommendation system for ideologically congenial, extreme, and problematic recommendations. *Proceedings of the National Academy of Sciences*, 120(50).
- [4] Hussein, E., et al. (2020). Measuring misinformation in video search platforms: An audit study on YouTube. *Proc. ACM Hum.-Comput. Interact.*, 4(CSCW1).
- [5] Bouchaud, P., et al. (2024). Auditing the audits: Evaluating methodologies for social media platform audits. *Applied Network Science*, 9, 55.
- [6] Habib, H. (2025). Auditing algorithmic bias with emotionally-agentic sock puppets. *arXiv preprint arXiv:2501.15048*.
- [7] Lam, M. S., et al. (2023). 360 sociotechnical audits. *Proc. ACM Hum.-Comput. Interact.*, 7(CSCW2).
- [8] Papadamou, K., et al. (2020). Disturbed YouTube for kids: Characterizing and detecting inappropriate videos targeting young children. *ICWSM*, 14, 522–533.
- [9] Bridle, J. (2017). Something is wrong on the internet. *Medium*.
- [10] Tahir, R., et al. (2019). Bringing the kid back into YouTube Kids. *ASONAM*.
- [11] Clark, D. R., & Jno-Charles, A. B. (2025). The child labor in social media: Kidfluencers, ethics of care, and exploitation. *Journal of Business Ethics*.
- [12] Bakioğlu, A. (2024). Digital capitalism and child labor exploitation on YouTube. *Sociology Lens*.
- [13] Divon, T., et al. (2025). Children as concealed commodities. *New Media & Society*.
- [14] Anderson, J. (2025). Growing up online: Children, family vlogs, and the

monetization of childhood. [15] Laude, C. (2024). Family vlogging and child harm. *Jurimetrics*, 64(3). [16] Abidin, C. (2015). Micromicrocelebrity: Branding babies on the internet. *M/C Journal*, 18(5). [17] Steinberg, S. B. (2017). Sharenting: Children's privacy in the age of social media. *Emory Law Journal*, 66, 839. [18] Kopecky, K., et al. (2020). The phenomenon of sharenting and its risks in the online environment. *Children and Youth Services Review*, 119. [19] Keskin, A. D. (2023). Sharenting syndrome: An appropriate use of social media? *Cureus*, 15(5). [20] Ratner, A., et al. (2017). Snorkel: Rapid training data creation with weak supervision. *VLDB Endowment*, 11(3). [21] Ratner, A., et al. (2016). Data programming: Creating large training sets, quickly. *NeurIPS*, 29. [22] Bach, S. H., et al. (2019). Snorkel DryBell. *SIGMOD*. [23] Johnson, J. M., & Khoshgoftaar, T. M. (2022). A survey on classifying big data with label noise. *ACM Journal of Data and Information Quality*, 14(4). [24] Ma, H., et al. (2023). Adapting large language models for content moderation. *arXiv*. [25] Gilardi, F., et al. (2023). ChatGPT outperforms crowd workers for text-annotation tasks. *PNAS*, 120(30). [26] Törnberg, P. (2024). ChatGPT-4 outperforms experts and crowd workers in annotating political Twitter messages with zero-shot learning. *arXiv*. [27] Children's Online Privacy Protection Act (COPPA), 15 U.S.C. § § 6501–6506 (1998). [28] Kids Online Safety Act (KOSA), S. 1409, 118th Congress (2024). [29] Illinois Child Influencer Act, 820 ILCS 152 (2024). [30] Livingstone, S., & Stoilova, M. (2021). The 4Cs: Classifying online risk to children. [31] UNICEF. (2025). Trends in online platform regulation and children's rights. [32] Covington, P., et al. (2016). Deep neural networks for YouTube recommendations. *RecSys*. [33] Zhou, R., et al. (2010). The impact of YouTube recommendation system on video views. *IMC*. [34] Rieder, B., et al. (2018). From ranking algorithms to 'ranking cultures'. *Convergence*, 24(1). [35] Sandvig, C., et al. (2014). Auditing algorithms: Research methods for detecting discrimination on internet platforms. [36] Metaxa, D., et al. (2021). Auditing algorithms: Understanding algorithmic systems from the outside in. *Foundations and Trends in HCI*, 14(4).